

Nuclear Physics Angular Momentum Coupling Guide

This document provides the theoretical background and practical usage guide for the `angular_momentum_coupling.py` utility. This tool calculates allowed final nuclear states based on quantum mechanical selection rules.

1. Theoretical Basis

The code implements the **Channel Spin Coupling Scheme**, which is a standard method for determining angular momentum conservation in nuclear reactions.

Conservation Laws

In any nuclear reaction **Target + Particle → Final_State**, the following quantities are conserved:

- 1. **Total Angular Momentum (J):** $J_{\text{final}} = J_{\text{target}} + s_{\text{particle}} + \ell$
- 2. **Parity (π):** $\pi_{\text{final}} = \pi_{\text{target}} \times \pi_{\text{particle}} \times (-1)^\ell$

Where:

- **J_target:** Spin of the target nucleus.
- **s_particle:** Intrinsic spin of the interacting particle (projectile or transferred cluster).
- **ℓ :** Relative orbital angular momentum between the target and the particle.

Coupling Scheme Used in Code

The code performs the vector addition in two steps (Channel Spin representation):

- 1. **Calculate Channel Spin (S):** Couples the target spin and particle spin. $S = J_{\text{target}} + s_{\text{particle}}$ Possible values: $|J_{\text{target}} - s_{\text{particle}}| \leq S \leq J_{\text{target}} + s_{\text{particle}}$
- 2. **Calculate Final Spin (J_final):** Couples the channel spin with the orbital angular momentum. $J_{\text{final}} = S + \ell$ Possible values: $|S - \ell| \leq J_{\text{final}} \leq S + \ell$

2. Application to Different Reaction Types

To use the code correctly, you must identify the correct "Particle" input based on the reaction mechanism.

A. Resonance and Capture Reactions

Examples: (p, γ) , (n, γ) , (p, p')

In these reactions, a projectile fuses with the target to form a compound nucleus (resonance).

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **projectile**.

Reaction	Projectile	Particle Input
----------	------------	----------------

Reaction	Projectile	Particle Input
$^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$	Proton	1/2+
$^{12}\text{C}(n, \gamma)^{13}\text{C}$	Neutron	1/2+
$^{16}\text{O}(\alpha, \gamma)^{20}\text{Ne}$	Alpha	0+

B. Single-Nucleon Transfer Reactions

Examples: (d, p), (p, d), (^3He , d), (α , t)

In these reactions, a single nucleon is transferred between the projectile and the target. You must input the properties of the **transferred nucleon**, not the beam.

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **transferred nucleon** (n or p).

Reaction Type	Example	Transferred Object	Particle Input
Stripping	$^{16}\text{O}(d, p)^{17}\text{O}$	Neutron	1/2+
Pickup	$^{13}\text{C}(p, d)^{12}\text{C}$	Neutron	1/2+
Proton Stripping	$^{40}\text{Ca}(^3\text{He}, d)^{41}\text{Sc}$	Proton	1/2+

Note on j-transfer: In transfer physics, we often discuss the transferred total angular momentum j (where $\mathbf{j} = \mathbf{\ell} + \mathbf{s}$). The code lists results by Channel Spin S . The set of allowed final $J\pi$ values is identical regardless of the coupling order.

C. Two-Nucleon Transfer Reactions

Examples: (p, t), (t, p), (^3He , n), (α , d)

In these reactions, a pair of nucleons is transferred. The total spin of this pair depends on the coupling of the two nucleons.

1. Identical Nucleon Transfer (2n or 2p)

Reactions: (p, t), (t, p) (2n transfer); (^3He , n) (2p transfer)

For the transfer of two identical nucleons (2n or 2p) in the same shell model orbit, the Pauli Exclusion Principle dictates that they must be in a singlet state ($S=0$) if they are in a relative s-state (which is the dominant transfer mode).

- **ENSDF Policy:** For (p, t), (t, p), and (^3He , n) reactions, it is standard to assume the transferred pair is in an **anti-parallel spin state ($S=0$)**.
- **Particle Input:** 0+

2. Neutron-Proton Transfer (1n1p)

Reactions: (α , d), (d, α)

For the transfer of a neutron-proton pair (a deuteron), the Pauli principle does not restrict the spin to $S=0$. The deuteron itself has spin $S=1$.

- **Spin Configurations:** Both singlet ($S=0$) and triplet ($S=1$) transfers are possible.
- **Particle Input:**
 - If assuming singlet transfer: $0+$
 - If assuming triplet transfer (e.g., deuteron-like): $1+$
 - *Note: Selection rules often require analyzing both possibilities.*

Transfer Type	Transferred Pair	Spin Assumption	Particle Input
2n / 2p	Identical	Anti-parallel ($S=0$)	$0+$
1n1p	Non-identical	Parallel ($S=1$) or Anti-parallel ($S=0$)	$1+$ or $0+$

D. Cluster Transfer Reactions

Examples: (${}^6\text{Li}, d$), (${}^7\text{Li}, t$)

An alpha particle (or other cluster) is transferred.

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **transferred cluster**.

Reaction	Transferred Cluster	Particle Input
(${}^6\text{Li}, d$)	Alpha (${}^4\text{He}$)	$0+$

3. Interpreting the Output

When you run the code:

```
python angular_momentum_coupling.py 3/2- 1/2+
```

You receive output like:

L	Wave	Parity	s	Final J π
0	s	-	1	1-
0	s	-	2	2-

- **L (Wave):** The orbital angular momentum of the incoming/transferred particle.
 - $L=0$ (s-wave), $L=1$ (p-wave), etc.
- **Parity:** The parity of the final state, determined by $\pi_f = \pi_i \pi_p (-1)^L$.
- **s (Channel Spin):** The intermediate coupling of Target + Particle.
 - For a $3/2^-$ target and $1/2^+$ particle, spins can align ($3/2+1/2=2$) or anti-align ($3/2-1/2=1$).
- **Final J π :** The allowed total angular momentum and parity of the final state.

Example Analysis: $^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$

- **Target:** ^{59}Cu (3/2-)
- **Particle:** Proton (1/2+)
- **Result:**
 - If capture occurs via **s-wave** ($L=0$):
 - Channel spins $S=1, 2$.
 - Final states $J\pi = 1-, 2-$.
 - If capture occurs via **p-wave** ($L=1$):
 - Parity flips (- to +).
 - Final states include $0+, 1+, 2+, 3+$.